

Tutorial: Building Reproducible Wi-Fi RF Fingerprinting Pipelines: Signal Collection, Datasets, and Evaluation

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Abstract

This tutorial provides a hands-on introduction to reproducible Wi-Fi radio frequency fingerprinting (RFF) pipelines for IoT device identification. Recent RFF research has demonstrated promising identification performance, yet reproducibility remains challenging due to inconsistent signal collection procedures, fragmented datasets, and non-standardized evaluation methodologies. This tutorial introduces an end-to-end experimental workflow covering Wi-Fi preamble data collection, RF signal preprocessing, feature extraction, baseline model reproduction, feature-separability analysis, and evaluation design. Using SMoRFFI as the primary case study, participants will work with large-scale Wi-Fi measurements collected from 123 same-model IoT devices, including both raw I/Q preambles and extracted RF features. The tutorial also demonstrates how to use our signal-collection software to acquire reproducible IEEE 802.11g preamble records. Accompanying executable notebooks and experimental resources will enable participants to reproduce the complete RFF pipeline. The session is designed for researchers interested in ubiquitous sensing, wireless device intelligence, IoT systems, and reproducible experimental practice.

CCS Concepts

• **Security and privacy** → **Authentication**; • **Human-centered computing** → **Ubiquitous and mobile computing theory, concepts and paradigms**; • **Hardware** → **Wireless devices**.

Keywords

RF fingerprinting, Wi-Fi device identification, IoT sensing, reproducible benchmark

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1 Introduction

Ubiquitous systems increasingly depend on dense wireless and IoT deployments, including smart environments, wearable devices, edge sensing platforms, and cyber-physical systems. In these settings, device identity affects data provenance, access control, and trust in sensed information. Software-level identifiers such as MAC addresses can be spoofed, randomized, or reused. Since RFF exploits transmitter-dependent hardware imperfections reflected in RF signals to distinguish devices, it provides a practical physical-layer approach for IoT device identification [4, 9].

However, turning RFF into a reproducible research workflow remains challenging for many researchers. Practical RFF experiments require knowledge of software-defined radio (SDR) operation, Wi-Fi protocol structure, synchronization, RF hardware impairments, feature construction, dataset organization, and evaluation protocol design [8, 11]. These components are often scattered across wireless communication, signal processing, machine learning, and system experimentation. As a result, newcomers face a high setup cost to run a meaningful RFF experiment.

Although existing studies have produced useful datasets, RF features, and models, the community still lacks compact educational workflows for reproducible RFF research [1, 7]. Collection programs are not always released, preprocessing choices can be under-specified, feature definitions vary across studies, and evaluation protocols differ across datasets and papers. This fragmentation limits reproducibility and makes it difficult for UbiComp/ISWC researchers who want to connect wireless identity with real-world sensing deployments.

Same-model device identification provides a rigorous educational sandbox for this tutorial. In this setting, devices share the same vendor and hardware model, so participants cannot rely on easy model-level differences. They have to inspect signal quality, understand feature behavior, examine model errors, and evaluate how data splits and sample budgets affect reported performance. This setting is therefore useful for teaching both the technical basis of RFF and the evaluation discipline needed for reproducible device-identification studies.

Motivated by these observations, this tutorial provides a hands-on path for building reproducible Wi-Fi RFF pipelines. Participants will first learn the Wi-Fi preamble collection process through an instructor-led acquisition walkthrough based on SDR, GNU Radio, Wi-Fi access points, and IoT transmitters. They will then use

open Wi-Fi preamble records, extracted RF features, and executable notebooks to inspect data, reproduce baseline identification results, diagnose feature separability, and examine evaluation protocols. The hands-on labs use open resources from SMoRFFI as the experimental sandbox, which includes data from 123 commercial IoT devices and provides a reproducible framework covering data collection, feature extraction, and benchmark evaluation [6].

After completing the tutorial, participants will be able to build, run, debug, and adapt a reproducible Wi-Fi RFF workflow. They will learn how to inspect preamble records, interpret RF features such as carrier frequency offset, phase error, magnitude error, I/Q gain imbalance, and fractal dimension [2, 3, 5, 9, 10, 12], reproduce baseline identification experiments, diagnose feature importance and confusion patterns, and evaluate the impact of data splits, device subsets, and sample budgets. All participant-facing hands-on activities will run on prepared data and executable notebooks, while the signal collection workflow will be introduced through demonstration materials and optional hardware display. This tutorial aims to help the UbiComp/ISWC community develop practical skills for reproducible wireless device-identification research.

2 Tutorial Overview

This tutorial is designed as a guided lab for RFF-based identification. Its goal is to help participants complete a workflow that usually requires separate preparation in wireless setup, data cleaning, feature interpretation, and benchmark implementation. As illustrated in Figure 1, the end-to-end RFF pipeline includes wireless signal acquisition, feature construction, recognition and decision, and evaluation with deployment considerations.

The acquisition stage will be demonstrated on site by the instructors using the USRP B210 receiver, GNU Radio environment, Wi-Fi access point, and M5Stack transmitters. Participants will observe the process from packet transmission to Wi-Fi preamble capture and record generation. The executable hands-on labs will then use prepared Wi-Fi records and notebooks, allowing participants to inspect signal records, construct and interpret RF features, reproduce baseline identification results, and examine evaluation choices under stable session conditions. The final part of the session will be used for output checking, discussion, and guidance on adapting the workflow to participants' own datasets.

3 Tutorial Structure and Learning Outcomes

As shown in Table 1, the tutorial is planned as a half-day session. Its structure follows the experimental workflow introduced in the previous section, moving from background concepts to executable exercises and then to evaluation design. Each module combines concise technical explanation with practical activities, so participants can build a reproducible RFF benchmark workflow during the session. The final discussion will help participants connect the released workflow to their own datasets and identify open challenges in reproducible Wi-Fi RFF research.

By the end of the tutorial, participants will be able to:

- Understand the Wi-Fi RFF data acquisition process, including packet transmission, preamble capture, and record generation;

- Build a working RFF-based identification pipeline from prepared Wi-Fi preamble and feature records;
- Compute selected RF features and visualize feature distributions for device-level fingerprint analysis;
- Train a baseline IoT device-identification model and inspect accuracy, feature importance, and confusion patterns;
- Debug common implementation issues in RFF experiments, including data loading errors, feature-format mismatches, and inconsistent model outputs;
- Apply the released workflow to their own wireless sensing or physical-layer device identification datasets.

4 Target Audience

4.1 Participant Profile

This tutorial is designed for three groups of UbiComp/ISWC participants. The first group includes researchers working on ubiquitous sensing, IoT systems, mobile computing, and wearable computing who need device-level trust, data provenance, or wireless identity mechanisms in real-world deployments. These participants may be familiar with sensing pipelines and machine-learning evaluation, but may lack practical experience with Wi-Fi preamble data, RF impairments, and physical-layer identification workflows.

The second group includes students and early-stage researchers who want to enter RFF-based device identification. For this group, the main barrier is the lack of a compact and executable path from raw wireless records to a working identification pipeline. The tutorial addresses this barrier through guided notebooks, a prepared dataset subset, and instructor-supported debugging.

The third group includes wireless-sensing, IoT security, and edge-intelligence researchers who already have experience with machine learning or wireless systems, but want to learn reproducible benchmark design for device identification. These participants will benefit from the evaluation modules on feature separability, data leakage, split design, and reproducible reporting.

4.2 Prerequisites

Participants are expected to have basic Python programming experience and basic familiarity with supervised machine learning. No prior experience with software-defined radios, GNU Radio, IEEE 802.11 signal processing, or RFF is required.

For local execution, participants should install Python 3.10 or later, JupyterLab or Jupyter Notebook, NumPy, pandas, scikit-learn, and Matplotlib. A tested environment file will be provided before the tutorial. Participants may also use a browser-based cloud notebook environment prepared by the organizers, which will reduce installation barriers and serve as a backup option.

Docker is not required for the hands-on identification labs. It will only be used to demonstrate the original data-acquisition environment. Participants who want to reproduce the acquisition workflow after the conference may optionally install Docker Desktop or a Linux-based Docker environment.

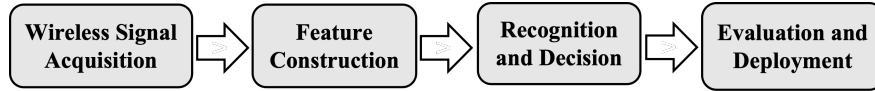


Figure 1: A simplified end-to-end RFF pipeline.

Table 1: Planned half-day tutorial structure.

Time	Module	Activity and Skill Outcome
0–15 min	Task Introduction	Define the target task: RFF-based identification for IoT devices.
15–40 min	Signal and Feature Introduction	Introduce the Wi-Fi preamble and RF feature knowledge needed for the lab.
40–60 min	Data Acquisition Pipeline Walkthrough	Instructors present a data acquisition pipeline with USRP B210, GNU Radio and M5Stack, then explain the data-collection process of the released Wi-Fi records.
60–90 min	Lab 1: Load and Explore the Dataset	Participants load the prepared Wi-Fi RFF dataset subset, explore device labels and preamble records, and identify RF feature fields.
90–110 min	Coffee Break and Hardware Display	Participants can inspect the USRP B210, M5Stack transmitter, and acquisition setup materials.
110–140 min	Lab 2: Construct and Visualize RF Features	Participants compute selected RF features and visualize feature distributions.
140–170 min	Lab 3: Build and Evaluate the Baseline	Participants build and run a device-identification baseline using prepared notebooks. They train a baseline classifier, generate identification outputs, and inspect accuracy and confusion patterns.
170–190 min	Debug and Output Checking	Participants compare their intermediate results with instructor-provided reference outputs. Instructors help resolve setup and implementation issues.
190–210 min	Wrap-up, Workflow Adaptation, and Open Discussion	Recap the hands-on workflow, discuss adaptation to participants' own datasets, and identify open challenges in reproducible RFF research.

4.3 Expected Attendance

We expect 25–40 participants. This estimate is based on the overlap between the tutorial topic and several active UbiComp/ISWC communities, including ubiquitous sensing, IoT infrastructure, wearable and mobile systems, edge intelligence, wireless sensing, and trustworthy connected-device deployment. The tutorial is also aligned with the conference interest in hands-on training, software demonstrations, and practical methods for emerging ubiquitous-computing topics.

We plan to keep the session within a manageable size for instructor support. If the number of registered participants exceeds the expected range, we will rely on cloud notebooks, pre-tutorial setup checks, and support from the instructional team. For a fully instructor-supported local setup, we recommend a soft limit of 40 participants.

5 Tutorial Materials and Logistics

5.1 Tutorial Materials

The tutorial materials will include slides, a Wi-Fi RFF dataset drawn from SMoRFFI, executable Jupyter notebooks, setup instructions, reference outputs, and supporting materials for the instructor-led acquisition demonstration. The dataset subset will contain representative Wi-Fi records, including device labels, preamble-related fields, and extracted RF features. The notebooks will support the main hands-on labs: record inspection, RF feature construction and visualization, baseline identification, and result diagnosis. Each

notebook will include checkpoints and reference outputs so that participants can verify intermediate results during the session.

5.2 Hardware, Software, and Licenses

The live hands-on labs require only a laptop and a modern web browser when participants use the prepared cloud notebook environment. Local execution will use Python 3.10 or later, Jupyter, NumPy, pandas, scikit-learn, and Matplotlib. The environment file and installation instructions will be released before the tutorial. As shown in Figure 2, the experimental platform and its acquisition workflow, including the USRP B210 receiver, GNU Radio environment, Wi-Fi access point, and M5Stack transmitters, will be presented through an instructor-led on-site walkthrough.

The notebooks and supporting scripts will be released under the MIT License. Slides will be released for educational reuse with attribution. The tutorial dataset subset will follow the license terms of the released SMoRFFI dataset. The Docker image or Dockerfile for the acquisition demonstration will be made publicly accessible when redistribution of all included components is permitted. If any third-party component imposes a stricter license, the release package will clearly document the corresponding restriction.

5.3 Ethical and Privacy Considerations

The tutorial dataset does not contain human-subject data, personal identity information, user behavior logs, or application-layer communication contents. The device labels correspond to laboratory-owned IoT devices used for controlled RFF experiments. Any on-site hardware interaction will use organizer-provided M5Stack devices,

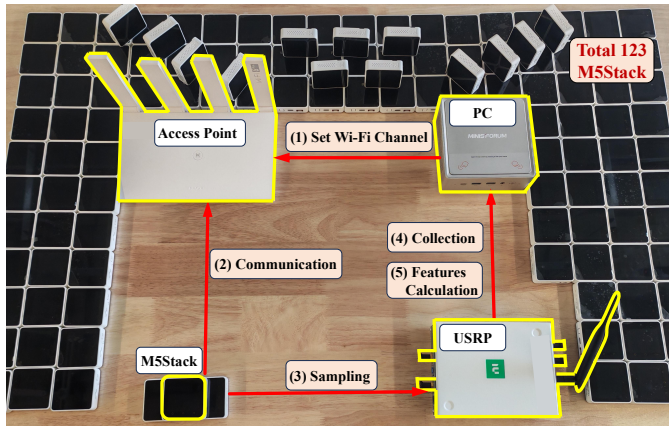


Figure 2: The wireless signal acquisition workflow.

and no personal wireless devices will be recorded during the session. The tutorial focuses on physical-layer device identification and reproducible benchmark design. During the session, we will also discuss responsible use of RFF, including the need to respect local radio regulations, avoid unauthorized device tracking, and separate controlled research datasets from real-world identity-sensitive deployments.

6 Instructional Team

The tutorial will be delivered by instructors with complementary expertise in Wi-Fi RFF, SDR-based acquisition, open dataset construction, and reproducible benchmark evaluation. The instructional team has experience in building RFF datasets, implementing feature-extraction pipelines, designing benchmark evaluations, and preparing executable teaching materials. At least two organizers will participate on site as instructors, with additional support for notebook-based labs and hardware demonstrations.

Jinxiao Zhu is an Assistant Professor at Tokyo Denki University. Her research focuses on physical-layer security and RFF. She will serve as the lead instructor for the RFF and physical-layer identification modules. She will also guide discussions on RF data collection, RF feature interpretation, and practical considerations for reproducible RF fingerprinting experiments.

Zhen Jia is an Assistant Professor in the Faculty of Engineering at Reitaku University. His research focuses on RFF and physical-layer device identification. He will support the instructor-led hardware walkthrough during the hands-on labs.

Wenhao Huang is an Assistant Professor at Keio University. His research interests include pedestrian sensing, wireless sensing, and applied machine learning for mobile and ubiquitous systems. He will support the hands-on notebook activities and participant discussions.

Zewei Guo is a doctoral student at Future University Hakodate. His research interests include RFF, wireless covert communication, and satellite communications. He will support the dataset walkthrough during the hands-on labs.

Yin Chen is an Associate Professor in the Faculty of Engineering at Reitaku University. He has extensive research and teaching

experience in RFF, wireless network security, machine learning, and related applications. He will coordinate the tutorial organization and instructional planning.

7 Conclusion

This tutorial offers a compact and hands-on path to reproducible RFF for IoT device identification. Using an open Wi-Fi RFF dataset, executable notebooks, and an instructor-led SDR acquisition walkthrough, participants will build a workflow covering Wi-Fi preamble inspection, RF feature interpretation, baseline reproduction, model training, and performance evaluation. The session emphasizes practical skills for connecting signal collection, dataset organization, feature construction, and benchmark evaluation. We hope this tutorial will help participants conduct RFF experiments through reproducible workflows, interpretable features, and sound evaluation practices, thereby bridging the gap between RFF theory and practical deployment in IoT and ubiquitous sensing applications.

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